

FINAL EXPLANATION: CHEMISTRY GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CHEMISTRY GRADE 10: ACIDS AND BASES (SUB-STRAND 3.1)

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

In this Final Explanation you answer the driving question: Does the alkalinity of a substance determine its uses in day to day life? Why are patients with hyperacidity issues given antacid medications? Use the evidence you collected in Lessons 1 to 10 and your final particle model (H^+ (aq) and OH^- (aq) ions). Write in full sentences in the space under each prompt. In every answer, give a claim, evidence (use numbers and observations) and reasoning (use particles and ions). Use the pH data below. Refer to the Kisumu products, Mr Kamau in Kericho and the pharmacist in Nairobi where you can.

Data table (pH measured with a pH meter at 25 °C):

Substance	pH
Lemon juice (Kisumu market)	2.0
Vinegar (5% ethanoic acid)	2.8
Pure water	7.0
Baking soda solution (as used for mandazi)	8.3
Bar soap solution	10.0

Part 1: Sorting products that look alike

Lemon juice, vinegar, baking soda solution and soap solution look similar in clear cups, yet Otieno found lemon sour and Wanjiru found soap slippery. Using the pH table and the indicator results from your tests (litmus, red cabbage extract and universal indicator),

Claim: Lemon juice and vinegar are acids, baking soda and soap are bases, and water is neutral. Evidence: Lemon juice (pH 2.0) and vinegar (pH 2.8) are below 7 and turn blue litmus red. Baking soda (pH 8.3) and soap (pH 10.0) are above 7 and turn red litmus blue. Water (pH 7.0) changes neither litmus paper. Red cabbage extract was red/pink in lemon juice, pink-red in vinegar, purple in water, blue-green in baking soda and green-yellow in soap. Universal indicator was red-orange (pH 2) in lemon juice, orange (pH 3) in vinegar, green (7) in water, blue (8) in baking soda and blue-purple (10) in soap. Reasoning: Different indicators have different colours, but each changes colour according to whether the substance is acidic, neutral or basic, so all give the same sorting. Universal indicator also shows how acidic or basic a substance is. Lemon juice (2.0) is more acidic than vinegar (2.8), a difference of 0.8 pH units. Soap (10.0) is more basic than baking soda (8.3), a difference of 1.7 pH units.

classify each product as acid, base or neutral. Explain how the indicators and pH values agree even though the colours differ. Which product in each pair is more acidic or more basic: lemon juice and vinegar, soap and baking soda? Give the pH evidence.

Part 2: What is inside acids and alkalis?

Explain at the particle level why lemon juice is sour and has a low pH, and why soap feels slippery. Use the Arrhenius definitions of an acid and an alkali, and the concentration data: lemon juice $H^+ = 1 \times 10^{-2}$ mol/L, vinegar H^+ about 1.6×10^{-3} mol/L, water 1×10^{-7} mol/L, soap $OH^- = 1 \times 10^{-4}$ mol/L. Explain how much more H^+ lemon juice has than vinegar, and say how you know. Describe how the Model you drew in Lesson 1 changed.

Claim: Lemon juice and vinegar are sour with low pH because they release H^+ (aq) ions in water, and soap is slippery because it is an alkali that produces OH^- (aq) ions. Evidence: Lemon juice has H^+ of 1×10^{-2} mol/L and vinegar about 1.6×10^{-3} mol/L, far more than pure water at 1×10^{-7} mol/L. Soap has OH^- of 1×10^{-4} mol/L, which is 1000 times more than water (1×10^{-7} mol/L). Reasoning: An acid is a substance that produces H^+ (aq) ions in water, so the citric acid in lemon juice and the ethanoic acid (acetic acid) in vinegar release H^+ (aq). A lower pH means a higher H^+ concentration, and each pH unit is a 10-fold change. Lemon juice is 0.8 pH units lower than vinegar, so it has about 6 times more H^+ ($10^{0.8}$ is about 6.3), which is why it is more sour. An alkali is a base that dissolves in water and produces OH^- (aq) ions; these ions react with the oils on the skin, giving the slippery feel. Baking soda gives fewer OH^- ions than soap, so it is only weakly basic (pH 8.3). My Lesson 1 model had only a guess that something differed inside the products; my final model shows more H^+ (aq) in lemon juice and vinegar and more OH^- (aq) in baking soda and soap.

Part 3: How acids and bases behave

When vinegar was added to baking soda it fizzed. Magnesium ribbon was placed in dilute hydrochloric acid. Describe what you observed and how you tested each gas. Explain why these reactions happen with acids but not with soap or water, and write word equations for the two reactions. Then explain what happens, in terms of ions, when an acid and a base are mixed.

Claim: Acids react with carbonates and reactive metals because of their H^+ (aq) ions, and acids and bases react together in neutralisation. Evidence: Vinegar and lemon juice fizzed with baking soda, and the gas turned limewater milky, showing carbon dioxide; water with baking soda did not fizz. Magnesium in dilute hydrochloric acid gave bubbles and the gas burned with a squeaky pop (lighted splint), showing hydrogen; magnesium in water gave no bubbles. Reasoning: The H^+ (aq) ions in acids react with the carbonate-type baking soda and with magnesium. Acid + carbonate gives salt + water + carbon dioxide. Acid + metal gives salt + hydrogen. Soap and water do not release H^+ (aq) so they do not do this. In neutralisation, acid + base gives salt + water; in terms of ions, H^+ (aq) + OH^- (aq) gives H_2O (l). In the investigation, universal indicator moved from red towards orange, yellow and green as magnesium hydroxide or baking soda was added to dilute hydrochloric acid, showing that the pH rose.

Part 4: Strength, safety and uses

Otieno asked: if both are acids, why is lemon juice lower on the graph than vinegar? Use the data for 0.1 M hydrochloric acid (pH 1), 0.1 M ethanoic acid (pH 3), 0.1 M sodium hydroxide (pH 13) and 0.1 M ammonia solution (pH 11) to explain the difference between strength and concentration. Then explain why a weak base such as magnesium hydroxide is a safer antacid than a strong base, and why soap is useful for cleaning but harmful if swallowed.	Claim: Strength depends on how much an acid or base ionises, not on how much is dissolved, and strength affects safety and uses. Evidence: At the same concentration of 0.1 M, hydrochloric acid has pH 1 but ethanoic acid has pH 3, and sodium hydroxide has pH 13 but ammonia has pH 11. Reasoning: Hydrochloric acid and sodium hydroxide are strong because they ionise almost completely, giving many H^+ (aq) or OH^- (aq) ions. Ethanoic acid and ammonia are weak because they ionise only partly, giving fewer ions even though the same amount is dissolved. Concentration is how much is dissolved; strength is how much ionises. A strong base would add a lot of OH^- (aq) and could be harmful in the body, so a weak base such as magnesium hydroxide is a safer antacid. Soap (pH 10.0) has OH^- (aq) that helps remove grease, which makes it useful for cleaning, but the same alkalinity is harmful if swallowed, so it needs safety precautions. In the laboratory we never taste chemicals and we wear goggles.
--	--

Part 5: Answering the driving question

Write your final Claim-Evidence-Reasoning answer to the driving question: Does the alkalinity of a substance determine its uses in day to day life? Why are patients with hyperacidity issues given antacid medications? Use at least three uses (for example vinegar, soap, baking soda, Mr Kamau's lime in Kericho, the pharmacist in Nairobi). Include the stomach pH values and the soil pH values. Say what else, besides alkalinity, matters.	Claim: Yes, alkalinity largely determines uses, together with strength and safety. Patients with hyperacidity are given antacids because the base in the antacid neutralises excess stomach hydrochloric acid. Evidence: Vinegar (pH 2.8) is an acid used by mama mboga for pickling kachumbari and onions. Soap (pH 10.0) is an alkali used for cleaning grease. Baking soda (pH 8.3) is weakly basic and safe in small amounts in mandazi. In Kericho, Mr Kamau uses lime to raise acidic soil from pH about 4.5 to about 6.0 for maize. After eating spicy nyama choma in Nairobi, a patient with hyperacidity is given magnesium hydroxide, which raises stomach pH from about 1.5 to 2 to about 3.5 to 5. Reasoning: Bases neutralise acids: acid + base gives salt + water, and H^+ (aq) + OH^- (aq) gives H_2O (l). The OH^- (aq) from lime reacts with H^+ (aq) in the soil, and the base in the antacid reacts with H^+ (aq) from stomach hydrochloric acid, raising the pH to a safer level. Acids are useful for preservation and flavour, and bases for cleaning, baking and neutralising acids. Strength and safety also matter: a weak base such as magnesium hydroxide is chosen as an antacid because it is safer than a strong base, and soap is useful but harmful if swallowed.
--	--

RUBRIC

Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim answers the driving question	Gives a clear claim that alkalinity, with strength and safety, largely determines uses, and states why antacids are given for hyperacidity.	Gives a claim about uses and antacids but leaves out strength or safety.	Claim is missing, unclear or does not address both parts of the question.

Use of evidence and data	Uses exact pH values, concentrations, indicator colours, reaction tests and stomach/soil pH changes correctly in every part.	Uses most data correctly, with one or two missing or small errors.	Uses little data, or data is incorrect or not linked to the claim.
Particle-level reasoning	Explains using H^+ (aq) and OH^- (aq), the Arrhenius definitions, the 10-fold pH change and H^+ (aq) + OH^- (aq) gives H_2O (l) accurately.	Mentions H^+ and OH^- ions correctly but explanations are incomplete.	Does not use ions, or confuses acids and bases.
Strength, concentration and safety	Clearly separates strength from concentration using the HCl, ethanoic acid, NaOH and ammonia data, and links this to safe antacid and soap use.	Describes strength correctly but does not link to safety clearly.	Confuses strength with concentration or omits safety.
Kenyan context and communication	Links to Kisumu products, Kericho lime and the Nairobi pharmacist; uses correct terms in clear, organised sentences.	Uses some contexts and mostly correct terms.	Little context; terms are misused or writing is hard to follow.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

Licensing & Copyright: © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.